10-30 第二篇

Geothermal Energy: Power from the Earth

The center of the Earth is extremely hot. This heat rises toward the surface in some places, warming underground water and creating steam. Geothermal energy is the power we get from this heat—a clean, reliable source that has been used for thousands of years for bathing and heating, and more recently for generating electricity. [1] Geothermal energy offers a steady, nearly endless supply of power that does not depend on the weather or the time of day, unlike solar or wind energy.

Not every place on Earth has easy access to geothermal resources. The best sites are often near volcanic areas, fault lines, or places where the Earth's crust is thin. Iceland, for example, uses geothermal energy for most of its heating and a large share of its electricity. Other countries with strong geothermal potential include New Zealand, Indonesia, and the United States. [2] Where geothermal resources exist, they can provide a large share of local energy needs, reducing reliance on fossil fuels and cutting carbon emissions.

Geothermal power plants work by drilling deep wells to reach hot water or steam. The steam turns turbines, which generate electricity. The used water is often pumped back underground, so the resource can be reused. Unlike coal or gas plants, geothermal plants produce very little air pollution and no direct greenhouse gas emissions from burning fuel. [3] Geothermal energy is one of the cleanest forms of power generation available today, making it an attractive option for countries seeking to reduce their environmental footprint.

There are some challenges. Drilling deep wells is expensive, and not all projects succeed in finding enough heat. In some areas, geothermal activity can cause small earthquakes or release gases that need to be managed. Scientists are working on new technologies to tap geothermal energy in more places, including areas that are not naturally hot. [4] As technology improves and costs fall, geothermal energy could play a larger role in the global transition to clean energy.

Geothermal energy will not replace all other energy sources, but it has an important part to play. For regions with the right conditions, it can provide stable, low-carbon power for decades. [5] Together with solar, wind, and other renewable sources, geothermal energy can help build a cleaner, more sustainable energy system for the future.

选项

A. Geothermal energy offers a steady, nearly endless supply of power that does not depend on the weather or the time of day, unlike solar or wind energy. B. Where

geothermal resources exist, they can provide a large share of local energy needs, reducing reliance on fossil fuels and cutting carbon emissions. C. Geothermal energy is one of the cleanest forms of power generation available today, making it an attractive option for countries seeking to reduce their environmental footprint. D. As technology improves and costs fall, geothermal energy could play a larger role in the global transition to clean energy. E. Together with solar, wind, and other renewable sources, geothermal energy can help build a cleaner, more sustainable energy system for the future. F. For example, the city of Reykjavik in Iceland heats nearly all its buildings using geothermal hot water from nearby wells.

答案键（Answer Key）

[1] A [2] B [3] C [4] D [5] E

深度解析

- **[1] A**：
  - **线索**：空前介绍地热能来源——地球内部热量、地下热水与蒸汽、清洁可靠、历史悠久，空后需概括地热能的独特优势。A的"steady, nearly endless supply"与"reliable"呼应，"does not depend on the weather or the time of day, unlike solar or wind"突出与太阳能、风能的对比，定义地热优势。
- **[2] B**：
  - **线索**：空前说地热资源分布不均，列举冰岛、新西兰、印尼、美国等有利地区，空后需说明有资源地区的利用价值。B的"Where geothermal resources exist"承接前文分布，"provide a large share of local energy needs"与冰岛等例对应，"reducing reliance on fossil fuels and cutting carbon emissions"与清洁能源主题衔接。
- **[3] C**：
  - **线索**：空前说地热电站原理——钻井取蒸汽、驱动涡轮、回灌再利用、几乎无空气污染和温室气体，空后需总结其环保地位。C的"one of the cleanest forms of power generation"与"very little air pollution and no direct greenhouse gas emissions"对应，"attractive option for countries seeking to reduce their environmental footprint"与前文环保特点呼应。
- **[4] D**：
  - **线索**：空前说挑战——钻井贵、成功率有限、可能诱发地震或释放气体、科学家研发新技术拓展应用，空后需展望技术发展带来的前景。D的"As technology improves and costs fall"与"new technologies... tap geothermal in more

places"呼应，"could play a larger role in the global transition to clean energy"是合理展望。

- **[5] E**：
  - **线索**：空前说地热不能替代所有能源，但在适宜地区可长期稳定供能，空后需总结与其他可再生能源的协同。E的"Together with solar, wind, and other renewable sources"与"will not replace all"形成递进，"help build a cleaner, more sustainable energy system"是全文收尾的合理升华。
- **干扰项分析**：
  - **F. For example, the city of Reykjavik in Iceland heats nearly all its buildings using geothermal hot water from nearby wells.**：以For example开头，为具体城市举例。各空均为概括或过渡句；[2]虽涉及冰岛，但需的是"有资源地区可满足大量需求"的概括，而非单一雷克雅未克案例。